

JESSE KIM

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Electrical Engineering, GPA: 4.00/4.00

Expected May 2028

Awards/Honors: Best Overall Award - URSA Research Symposium (1/23), James Scholar, Dean's List Award 2x

EXPERIENCE

GPU Research Engineering Intern

August 2026 - Current

Advised by Prof. Deborah Levin

- Implementing **GPU-direct CUDA-MPI** communication to eliminate host-stage transfers in a 3D **multi-GPU** octree PIC-MCC plasma simulation, reducing overhead in E-Forest-of-Trees z-boundary data exchange during Poisson solves

TPU Kernel Engineering Intern

August 2026 - Current

Systems Platform and Intelligence Lab - Prof. Jian Huang

- Porting a CUDA-based LLM serving engine (Qwen) to **Google TPUs** using **JAX/Pallas**, implementing optimized Page-dAttention kernels, continuous request batching, and KV-cache management to improve serving throughput

GPU Research Engineering Intern

May 2026 - Aug. 2026

Advised by Ph.D. Candidate Gyuha Lim

- Developed custom **CUDA** kernels in **C++17** to parallelize a 1D3V Particle-In-Cell simulation, accelerating the kinetic tracking of **3M+ particles** and eliminating primary bottlenecks
- Achieved **24.1x speedup** in particle push, **5.12x speedup** in charge deposition, and **1.38x speedup** in particle boundary conditions computations over single-threaded CPU code with **RTX 4060 GPU**
- Optimized memory throughput by leveraging Struct-of-Arrays data layout for coalesced memory access, and minimized latency by mitigating PCIe bus transfers and warp divergence

HPC Research Engineering Intern

January 2026 - May 2026

Advised by Ph.D. Candidate Gyuha Lim

- Designed and benchmarked various optimized 1D Poisson solvers in **C++17** for a 1D3V Particle-In-Cell simulation, evaluating stability, CPU time, and error convergence rates
- Tailored diverse numerical methods to a 1D solver matrix, including a Thomas Algorithm and 4 distinct preconditioned conjugate-gradient variants (Jacobi, SSOR, Incomplete Cholesky, First-Order Neumann series)
- Achieved a **651x speedup** in field-solver time using the Thomas Algorithm compared to standard Conjugate Gradient and a **13x speedup** through iterative SSOR-preconditioning with multidimensional scalability

PROJECTS

GEMM Kernel Optimization Series

github.com/jtkim00000/gemm-series

CUDA, C++, Kernel Optimization

- Progressively optimized General Matrix Multiplication Kernels in **CUDA**, tailored to an **RTX 4060 GPU**, validated and benchmarked throughput across various matrix dimensions, comparing each kernel's throughput against **cuBLAS**
- Implemented tiling, thread coarsening, memory coalescing/corner turning, and bank-conflict avoidance, achieving a peak throughput **5.81 TFLOPS** on RTX 4060, **5.95x speedup** from naive GEMM kernel, and **61.6%** of cuBLAS throughput

Pipelined RISC-V CPU

github.com/jtkim00000/riscv-cpu

Verilog RTL & Verification, GTKWave, Computer Architecture

- Built a custom Pipelined RISC-V CPU in Verilog RTL from scratch, implementing 37-instruction subset of RV32I ISA
- Designed custom CPU datapath, first implementing single-cycle CPU, then extending into a 5-stage pipelined CPU (IF/ID/EX/MEM/WB) with forwarding, pipeline stalling, and register file write-through for full hazard resolution
- Verified CPU functionality across all instruction types, hazard scenarios, and components using hand-assembled RV32I test programs, GTKWave waveform analysis, and isolated module-level testbenches

SKILLS

Programming & Hardware: CUDA, C/C++, JAX/Pallas (TPU), SystemVerilog & Verilog, Python, RISC-V

Tools: Vivado, GTKWave, KiCAD, Git/GitHub, Linux, Tecplot